

## ELECTROMAGNETIC WAVES

1. Choose the correct statement :
  - (1) Time-dependent electric and magnetic fields give rise to each other
  - (2) Time dependent electric field gives rise to magnetic field but vice-versa is not true
  - (3) Time dependent magnetic field gives rise to electric field but vice-versa is not true
  - (4) There is no dependence between time varying electric field and magnetic field
2. Which waves are suitable for Radar systems used in aircraft navigation ?
  - (1) Radio waves
  - (2) Micro waves
  - (3) Infrared waves
  - (4) Visible rays
3. Which waves are produced by red hot bodies ?
  - (1) Visible light
  - (2) Ultraviolet rays
  - (3) Infrared rays
  - (4) Microwaves
4. Snakes can detect which of the following electro magnetic waves ?
  - (1) X-rays
  - (2) Gamma rays
  - (3) Ultraviolet rays
  - (4) Infrared waves
5. Exposure to which radiation produces more melanin ; causing tanning of the skin :
  - (1) UV radiation
  - (2) Infrared rays
  - (3) X-rays
  - (4) Gamma rays
6. UV lamps are used to :
  - (1) As detectors in satellites
  - (2) See the world around us
  - (3) Kill germs in water purifiers
  - (4) Communicate through Cellular phones
7. Which rays can damage or destroy living tissues and organisms in case of unnecessary or over exposure :
  - (1) X-rays
  - (2) Cathode rays
  - (3) Ultraviolet rays
  - (4) Infrared waves
8. Which high frequency radiation is produced in nuclear reactions and also emitted by radioactive nuclei :
  - (1) Infrared rays
  - (2) Gamma rays
  - (3) Ultraviolet rays
  - (4) Microwave
9. EM waves can be produced by:
  - (1) Stationary charges
  - (2) Charges in uniform motion
  - (3) Accelerated charges
  - (4) Accelerated mass
10. If the total energy transferred to a surface in time  $t$  is  $U$ , then magnitude of total momentum delivered to a surface (for complete absorption) is given by :
  - (1)  $p = \frac{U}{C}$
  - (2)  $p = \frac{2U}{C}$
  - (3)  $p = \frac{U}{2C}$
  - (4)  $p = \frac{3U}{2C}$
11. Particle related to an electromagnetic wave is :
  - (1) Electron
  - (2) Proton
  - (3) Photon
  - (4) Phonon
12. Geiger tubes can detect which radiations :
  - (1) Microwaves
  - (2) Radiowaves
  - (3) Infrared rays
  - (4) Gamma radiations
13. EM waves used for studying crystal structure of solids :
  - (1) X-rays
  - (2) Microwaves
  - (3) Radio waves
  - (4)  $\gamma$ -rays
14. In a wave motion,
 
$$y = a \sin (kx - \omega t)$$
 $y$  can represent :
  - (1) Electric field
  - (2) Magnetic field
  - (3) Pressure
  - (4) All of these
15. Infrared radiations are detected by ;
  - (1) Photocell
  - (2) Pyrometer
  - (3) Monometer
  - (4) Photometer
16. If the frequency of oscillating particle is ' $n$ ', then the frequency of oscillation of its energy is :
  - (1)  $n$
  - (2)  $2n$
  - (3)  $\frac{n}{2}$
  - (4)  $4n$
17. If  $V_s$ ,  $V_x$  and  $V_m$  are the speed of gamma rays, X-rays and microwaves respectively in vacuum, then :
  - (1)  $V_s > V_x > V_m$
  - (2)  $V_s < V_x < V_m$
  - (3)  $V_s > V_m > V_x$
  - (4)  $V_s = V_x = V_m$

18. If sun radiates electromagnetic energy of  $3.9 \times 10^{26}$  watt and radius of sun is  $6.96 \times 10^8$  m then intensity of sunlight at the solar surface will be :-  
 (1)  $5.6 \times 10^6 \text{ W/m}^2$  (2)  $6.4 \times 10^7 \text{ W/m}^2$   
 (3)  $4.2 \times 10^6 \text{ W/m}^2$  (4)  $4.2 \times 10^7 \text{ W/m}^2$
19. The electric and magnetic field of an electromagnetic wave are :  
 (1) In phase and parallel to each other  
 (2) In opposite phase and perpendicular to each other  
 (3) In opposite phase and parallel to each other  
 (4) In phase and perpendicular to each other
20. A wave is propagating in a medium of electric dielectric constant 2 and relative magnetic permeability 50. The wave impedance of such a medium is :  
 (1)  $5 \Omega$  (2)  $376.6 \Omega$   
 (3)  $1883 \Omega$  (4)  $3776 \Omega$
21. A compass is placed between the plates of charging capacitor, then its deflection will :  
 (1) Be zero always  
 (2) will be non-zero always  
 (3) will become zero at steady state  
 (4) None
22. A parallel plate capacitor with plate area A and separation between plate d, is charged by a constant current I. Consider a plane surface of area  $\frac{A}{4}$  parallel to plates, what is displacement current through this area :  
 (1) I (2) 2I (3)  $\frac{I}{4}$  (4)  $\frac{I}{2}$
23. Water molecules rotate about own axis with frequency equivalent to that of :  
 (1) Radio waves (2) Micro waves  
 (3) Ultraviolet rays (4) None of these
24. Porcelain container remains unaffected and cool inside microwave oven because its molecules does not absorb radiation energy as its molecules vibrate and rotate with frequency :  
 (1) Higher than microwave frequency  
 (2) Lower than microwave frequency  
 (3) Same as microwave frequency  
 (4) Porcelain molecules are large so they don't vibrate and rotate
25. If antenna of portable AM radio is turned from vertical to horizontal position then signals are greatly diminished due to phenomenon of :  
 (1) Interference (2) Diffraction  
 (3) Polarization (4) Reflection
26. Heat waves are :  
 (1) Radio waves (2) X-rays  
 (3) Infrared (4) Ultraviolet
27. TV remote uses which radiations :  
 (1) Radio wave (2) Micro wave  
 (3) Infrared (4) UV
28. LASIK (Laser-assisted in situ keratomileusis) eye surgery uses which radiations :  
 (1) Microwave (2) Radiowaves  
 (3) UV rays (4)  $\gamma$ -radiations
29. Electromagnetic waves transport :  
 (1) Only energy  
 (2) Only momentum  
 (3) both energy and momentum  
 (4) neither energy nor momentum
30. Microwaves are produced by:  
 (1) Vibrations of atoms  
 (2) Radioactive decay  
 (3) Klystron valve  
 (4) Transition of electron from higher to lower energy level